

SNOWFLAKE COCO CLI HACKATHON · GCC EDITION

Zero to One Chain

Supply Chain Decision Twin: one governed truth, from raw ERP rows to an approved action

Team Name **Zero to One Chain**

Team Leader Name **Siddharth Shastri**

Team Size **1**

Problem Statement **Supply Chain Ontology & Governed Conversational Analytics**

Built end to end in Snowflake with Cortex Code (CoCo): dbt, Dynamic Tables, Tasks, Semantic View, Cortex Analyst, Cortex Search, Cortex Agents, Snowflake ML, Horizon governance, Streamlit

Same question, three different answers

Supply chain data lives in separate systems (ERP, TMS, SRM, IoT). Each team rebuilds its own logic, so Planning, Procurement and Logistics argue about whose number is right instead of acting on it.

Conflicting definitions

"On-time" means delivered lines to one team and all lines to another, so OTD differs by team.

No connected view

Nobody can trace supplier → part → customer order, so a supplier outage is noticed only when customers complain.

Answers nobody trusts

Chat-based BI hallucinates metrics; dirty source rows silently flow into dashboards.

Insight without action

Dashboards stop at a chart; decisions and approvals happen in email with no audit trail.

WHAT JUDGES CAN VERIFY

544K

raw rows across 11 source tables (22-38 columns each)

6 / 6

canonical metrics identical for all three personas

3

fully separate environments: DEV, UAT, PROD

0

manual steps from a DEV change to a verified UAT build

A decision twin: ontology, governed semantics, agents, action

1 Connect

Ontology of suppliers, parts, plants, customers, orders and shipments, with entity resolution across ERP and SRM supplier records.

2 Govern

One semantic view holds canonical metrics (OTD, OTIF, fill rate, landed cost, days of inventory). Masking, row access, tags and data metric functions are enforced in Snowflake.

3 Ask

A Cortex Agent answers in plain English with Cortex Analyst (24 verified queries) plus Cortex Search over 200 contracts and SOPs, always on governed definitions.

4 Act

Disruption simulator, data-driven recommendations, human approval and an audit trail. Delivered through a release train from DEV to UAT to PROD.

Result: every persona, dashboard and agent reads the same governed definition, and every recommended action is traceable from the raw row that triggered it to the person who approved it.

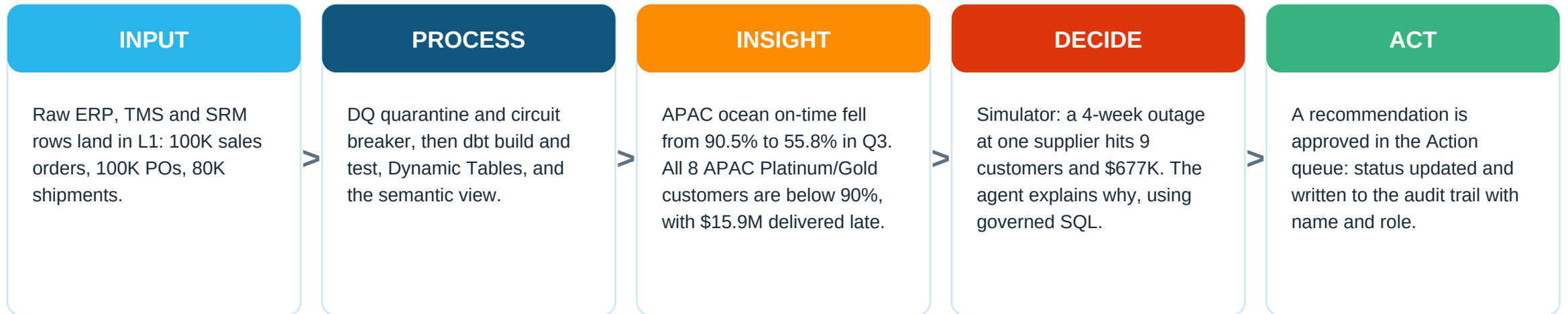
9-layer architecture, all native Snowflake

L1 Ingestion	11 raw source tables, 544K rows
L2 Stage	dbt staging views, typed and cleaned
L3 Validate	DQ quarantine, circuit breaker, DMFs
L4 Harmonize	Dynamic Tables: entity resolution
L5 Core	dbt marts: facts and dimensions (41 tests)
L6 Semantic	Semantic view, canonical metrics, consistency proof
L7 Intelligence	Cortex Agent, Analyst, Search, ML forecast + anomaly
L8 Action	Recommendations, approvals, alerts, release train
L9 Experience	Streamlit decision app, 8 pages, 3 personas

CROSS-CUTTING

- Governance: masking and row access policies, classification tags, 3 data metric functions
- Orchestration: hourly task DAG with circuit breaker and 3 alerts
- Environments: DEV, UAT, PROD as separate databases, each built by dbt
- Release train: change detection, auto UAT release, verified, human-approved PROD
- Cost control: resource monitor, 60s auto-suspend, one-click project stop

One working workflow, input to output



Every number above comes from the live app and the semantic view. The Cortex Agent returns the same figure because it generates SQL against the same governed metric (OTD = delivered lines only).

Conversational analytics you can trust

Same answer for every persona

- Semantic view SUPPLY_CHAIN_ANALYTICS: 4 logical tables, canonical metrics, 24 verified queries
- Cortex Agent SUPPLY_CHAIN_TWIN routes between Cortex Analyst (structured) and Cortex Search (200 contracts, SOPs, incident notes)
- Remediation agent proposes fixes for quarantined records; a human applies them
- Consistency proof task re-computes each metric as Planning, Procurement and Logistics every hour: 6 of 6 identical

Governed in Snowflake, not in the app

- Masking: supplier spend visible to Procurement only; contact emails masked for non-admin roles
- Row access: Logistics sees AMERICAS and EMEA customer rows only
- Classification tags (PII, FINANCIAL) on sensitive columns
- Data metric functions: negative values, out-of-range rates, orphan suppliers
- Quality gate: any critical issue blocks every promotion to UAT and PROD

Production-ready, not a notebook demo

41 / 41

dbt tests passing in DEV, UAT and PROD

14

Dynamic Tables keeping layers fresh

6

tasks in the hourly DAG, incl. release train

3

alerts: SLA breach, stockout, DQ critical

Release train: DEV > UAT (automatic) > PROD (human approval)

- At the end of each hourly DEV run, a content fingerprint (HASH_AGG of every raw table plus a hash of every view, procedure, Dynamic Table, task, semantic view, search service, agent and dbt project) is compared with the last release.
- Unchanged: logged and skipped, so an hourly release never wastes credits. Changed: DQ gate, data promotion, dbt build and tests, full-stack deploy to UAT, then UAT is verified (metric parity with DEV, 0 critical DQ issues, consistency proof).
- A verified build raises a PROD approval card. Approval is refused if DEV changed after verification, so PROD only ever receives exactly what passed UAT. Newer builds supersede older approvals.
- One switch starts or stops the whole project across all three environments; every action is audited.

Snowflake features, each chosen for a reason

dbt Projects on Snowflake

Versioned, tested transformations; the same code builds every environment

Dynamic Tables

Declarative freshness for harmonized layers without hand-written refresh jobs

Tasks + Alerts

Hourly DAG with circuit breaker; alerts raise recommendations automatically

Semantic View

Single home for metric definitions shared by the app, Analyst and the agent

Cortex Analyst + Agents

Natural-language questions answered with governed, inspectable SQL

Cortex Search

Grounds answers in contracts and SOPs, not only tables

Snowflake ML

OTD forecast and inventory anomaly detection inside the warehouse

Horizon governance

Masking, row access, tags and DMFs enforced at the data, for every tool

Streamlit in Snowflake

Decision app runs next to the data with the caller's governance

Resource monitor

Hard daily credit cap; the release train is change-driven to stay within it

Impact

1 truth

one governed definition per metric across app, agent and personas

Minutes

from a data change to a verified UAT build, with no manual steps

Seconds

to see which customers a supplier outage hits, before they call

100%

of decisions and releases audited with name, role and note

Next steps

- Connect live ERP/TMS feeds with Snowpipe Streaming or Openflow in place of the synthetic sources
- Add more domain models (finance, healthcare supply) on the same ontology and semantic layer
- Expose the agent in Snowflake Intelligence and Microsoft Teams for planners in the field
- Close the loop: write approved actions back to ERP purchase orders through external access